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Full Length Article

Influences of Taiwan's corporate social responsibility report management policy on the information transparency of its capital market

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Abstract

The management policy for Taiwan's corporate social responsibility (CSR) reporting was changed from elective to partial mandatory. Some companies that have already submitted their reports are also required to undergo assurance provided by accountants. The present study explores the influences of Taiwan's partial mandatory disclosure and partial mandatory assurance management policies for CSR reports on firms' investment to cash flow sensitivity which is used to measure the information transparency of the capital market. The results showed that investment to cash flow sensitivity decreased after mandatory disclosure and mandatory assurance. This implies that the partial mandatory disclosure and the partial mandatory assurance policies for CSR reports have a positive influence on mitigating information asymmetry and agency problems being experienced by firms. It also indicates that Taiwan's partial mandatory disclosure and partial mandatory assurance policies for CSR reports aid in increasing information transparency of its capital market.
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Keywords: Corporate social responsibility report; Mandatory disclosure; Mandatory assurance; Investment to cash flow sensitivity; Information transparency

1. Research motivation

The emerging capital market of Taiwan is either family-controlled or group-controlled. In an emerging capital market, information transparency is critical in attracting other sources of investment. In recent years, international standards related to corporate social responsibility (CSR) have emerged, which are actively being regulated by both the government and the market operators.

Taiwan's regulation regarding the preparation of CSR report was changed from elective to a partial mandatory requirement. The Financial Supervision and Management Commission (FSMC) issued the "Taiwan Stock Exchange Corporation Rules Governing the Preparation and Filing of Corporate Social Responsibility Reports by TWSE Listed Companies" in 2014. This stipulates that the listed companies belonging to industries concerning food, chemical, and financial and insurance, those that have not less than 50% of the revenue derived from food and beverage, and those with a capital stock that has achieved not less than NT$10 billion should refer to the Global Reporting Initiative's latest version of sustainability reporting guidelines, industry supplemental guidelines, and other applicable criteria based on industry characteristics when publishing their previous year's CSR report.

In addition to the partial mandatory disclosure management policy for CSR reports, the Taiwan authority also proposed to improve the quality of its partial mandatory assurance policy for CSR reports. Firms belonging to the food industry or those that have not less than 50% of revenue derived from food and beverage, should submit their reports to accountants for quality assurance. Because of Taiwan's relatively relaxed legal environment, the mandatory disclosure of CSR reports raises questions regarding their authenticity, which proves that these reports require assurance checking through reliable accountants. Aside from the enterprises' implementation and upholding

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of social responsibility, the Taiwan government also regards the CSR report as vital and necessary and deems it as an official document that has the same importance as the financial report.

Under the current CSR management policy, firms can be classified into three categories according to the nature of CSR submissions: (1) voluntary submission, (2) government-mandated submission, and (3) nondisclosure. Furthermore, these firms can also be classified according to the nature of their assurance source: (1) through an accountant, (2) through third-party assurers, and (3) no assurance done. In line with this, Taiwan's capital market is a potential setting for research concerning the impact of current CSR management policy on information transparency of its capital market. Taiwan's unique aspect of CSR management policy may be used as a benchmark by other countries regarding policy formulation and implementation. Further, the lack of literature analyzing the phenomena of this policy ultimately sets the driving force of this research.

This paper has two research goals: (1) to examine the effects of Taiwan's partial mandatory disclosure policy for CSR reports on the information transparency of its capital market; and (2) to investigate the influence of Taiwan's partial mandatory assurance policy for CSR reports on the information transparency of its capital market. To address these research goals, this paper used firms' investment-cash flow sensitivity to measure the information transparency of the capital market. The researcher aims to answer the following questions:

(1) Does the partial mandatory disclosure policy for CSR reports decrease firms' investment-cash flow sensitivity?
(2) Does the partial mandatory assurance policy for CSR reports decrease a firm's investment-cash flow sensitivity?

The other expected contributions of this paper are as follows.

First, this paper focuses on mandatory disclosure or assurance as opposed to voluntary disclosure or assurance which are the basis of current literature on CSR reporting. Companies voluntarily disclose CSR reports based on strategic considerations. Equilibrium in the disclosure strategy is a trade-off between the costs and benefits of disclosure. The earlier study tried its best to be as objective as possible; however, some limitations have been found which may have affected the empirical results. Among which is the research sample not being a perfectly random sample. Due to this, the observed research results may have been produced by a self-selection mechanism, which is not entirely causal. Therefore, the correlation between CSR disclosure and a firm's outcome documented in earlier studies may not also be necessarily causal. Unlike earlier studies, this paper investigated the economic consequence of mandatory CSR report disclosure, which avoided reverse causality because the mandatory disclosure regulation was unexpected and constituted an exogenous shock outside the firm. This means that the study's sample was more random because disclosure was mandatory rather than voluntary; thus, the statistical inferences did not suffer from reverse causality problem. It is, therefore, worthwhile to use Taiwan's setting to discuss CSR-related issues.

Second, this study explores the impact of Taiwan's mandatory CSR reports management policy on the information asymmetry of the capital market. The results of this paper can make incremental contributions to the literature in non-financial information-related fields.

Third, the present research provides managerial implications for policy formulation which can be used as a basis by the Financial Management Association in creating relevant regulations, for companies with unclear CSR policies, for the assurance service industry, and for the CSR management talents in higher education.

The succeeding sections of this paper are outlined as follows: the second part explores relevant literature surrounding the premise of the research problem; the third part describes the empirical methodology; the fourth part discusses the results of the research; and the final part recommends policy suggestions extrapolated from the research data.

2. Literature review and research hypotheses construction

2.1. CSR and information asymmetry

Cheng et al. (2014) pointed out that companies with better CSR performance reduced information asymmetry based on broader and more credible CSR disclosure practices. This implied that companies with better CSR performance are more likely to obtain funding with a lower financial constraint. Kim et al. (2012) believed that companies with efficient CSR policies will present financial information more responsibly by increasing its transparency and trustworthiness and decreasing its costliness and risk for data manipulation. They also found that it could decrease the chances of an external audit by the Securities and Exchange Commission.

Several studies found that CSR reports increased the availability and quality of data, and reduced information asymmetry between companies and investors (Chen et al., 2009; El Ghoul et al., 2011; Hail & Leuz, 2006; Khurana & Raman, 2004), leading to a reduction in capital constraints (Hubbard, 1998). Further, the empirical results of previous studies demonstrated that CSR reduced information asymmetry. Michaels and Grüning (2017) found that CSR disclosure was significantly and negatively associated with information asymmetry as well as the cost of capital. Using the firm-level CSR dataset of Australian publicly listed firms from 2004 to 2014, Nguyen et al. (2019) found that CSR performance was negatively associated with information asymmetry. Moreover, this negative relationship was stronger for larger firms and firms with stronger market power. Cui et al. (2018) also found an inverse association between CSR engagement and the proxies of information asymmetry using American-listed companies as samples.

2.2. CSR and agency problems

A company with an outstanding CSR performance, expressing its commitment and participation with a variety of

stakeholders, is based on mutual trust and cooperation (Jones, 1995). Jones (1995) pointed out that for commitment issues, ethical solutions are more efficient than mechanisms designed to contain opportunism; thus, companies that contract with stakeholders based on mutual trust and cooperation have reduced agency costs and transaction costs. This indicates that excellent CSR performance is related to better stakeholder participation, and limits the possibility of short-term opportunistic behavior (Benabou & Tirole, 2010).

Dhaliwal et al. (2011) found that companies with better CSR performance were more likely to disclose CSR activities to the markets, to highlight their long-term focus and differentiate them from other companies (Benabou & Tirole, 2010). Cheng et al. (2014) demonstrated that CSR reports established a positive feedback cycle. Initially, the CSR report improved the transparency of the company's social and environmental aspects and enhanced its governance structure; next, the CSR changed the internal control system which further enhanced the company's commitment, leading to the formulation of relevant regulations to increase the reliability of the report. Cheng et al. (2014) pointed out that companies with better CSR performance, based on more effective stakeholder participation, had better equity-related stipulations, which reduced agency costs and generated revenue/profit potential. This means that it is easier for companies with better social responsibility performance to obtain capital and to have lower financial constraints. In summary, CSR may help reduce agency problems.

2.3. CSR and investment to cash flow sensitivity

Regarding firms' investment activities, the perfect capital market theory, proposed by Modigliani and Miller (1958), pointed out that if a company has a project to invest in, it can obtain the funds it needs in the capital market. However, in the real market, many factors could affect the company's investment decisions and financing decisions related to each other. Several studies found that investment is affected by a company's internal cash flow. The theories included information asymmetry between company and capital market (Myers & Majluf, 1984) and agency issues between managers and shareholders (Jensen, 1986).

Since Fazzari et al. (1988) began to use the investment-cash flow sensitivity to explore financial constraints of corporate investment, it triggered a series of follow-up studies (such as Himmelberg & Petersen, 1994; Hoshi et al., 1991) to investigate the impacts of liquidity (internal cash flow) on investment. This issue continues to be a concern because the current practice is influenced by factors such as information asymmetry, transaction costs, agency problems, internal funds and external funds not being completely replaced; this leads to financial constraints creating a distortion effect on the allocation of corporate investment activities, which is harmful to value increase and affects long-term economic development.

Governments and capital market operators propose specific practices to improve information transparency of capital markets. In addition to the improvement of financial information, the promotion of non-financial information has become increasingly important. In recent years, an increasing trend of international norms related to CSR has emerged one after another. More and more countries are now actively encouraging or regulating companies to fulfill their social responsibilities. Taiwan's policy for CSR report preparation was changed from its original voluntary nature to mandatory requirements.

Based on the literature review, it can be assumed that CSR can reduce information asymmetry and agency problems which could further affect a company's external financing activities. These are explained in detail below:

CSR can reduce information asymmetry: companies that do their best in CSR tend to present their financial information more responsibly. Since CSR reports enhance information transparency, it can help reduce the information asymmetry between investors and companies as it is beneficial to companies' external financing; thus, the investment-cash flow sensitivity can be reduced.

CSR can reduce agency problems: Jensen (1986) proposed the free cash flow hypothesis based on the agency problem. For companies with insufficient regulatory mechanisms, management decisions may be driven by "empire building", leading managers to over-investment. The relationship between investment and cash flow may be positive in this case. Companies with better CSR performance are likely to have superior stakeholder participation; therefore, the short-term opportunistic behavior of company managers can be reduced which can decrease agency problems. Hence, companies with better CSR performance may have better corporate governance institutions. When corporate governance improves, agency problems are expected to decrease. Therefore, managers will not subject their cultural and personal biases towards CSR-based decisions, and the dependence of investment on cash flow will be reduced. Moreover, the reduction of agency problems facilitates the company's external financing, which could further reduce the investment-cash flow sensitivity.

Information asymmetry and agency problems tend to make companies rely on cash flow when investing. As reiterated above, CSR could help reduce information asymmetry and agency problems. Therefore, under the CSR report management policy, the mandatory disclosure will force firms to expose their CSR reports which could help alleviate information asymmetry and agency problems; this, in turn, could help firms decrease their investment-cash flow sensitivity.

In line with the findings of previous studies, this study proposes the following hypothesis:

H1. The mandatory disclosure management policy for CSR reports helps in alleviating the investment-cash flow sensitivity of companies that are forced to disclose their CSR reports.

Kawk and Choi (2015) studied the relationship between CSR and corporate financial constraints of 1584 companies in Korea between 2002 and 2011. Corporate financial constraints were measured by the KZ index, as proposed by Kaplan and Zingales (1997). The study found that there was a significant and negative correlation between the company's financial

constraints and the stakeholder-weighted CSR index, but there was no significant correlation between financial constraints and equal-weighted CSR index. This means that only the CSR strategy that stakeholders prioritize can have a positive impact on a company's ability to obtain capital market financing. Moradi et al. (2018) investigated the CSR, agency costs, and investment-cash flow sensitivity in listed companies on Tehran Stock Exchange. A sample of 120 companies from 2010 to 2016 was selected by systematic elimination sampling. The results showed that there was a negative and significant relationship between CSR and investment sensitivity to cash. This indicates that companies with a better CSR function tend to be less sensitive to investment. The above literature supports that firms with socially responsible practices are better positioned to obtain financing in the capital markets by reducing market frictions as well as agency costs.

The present study believes that Taiwan's CSR report management policy experience is worth discussing because of the following reasons:

First, the management policy of CSR report disclosure is different from other countries. Currently, the relevant literature on CSR and finance constraints is mostly under the background of voluntary disclosure (voluntary assurance) of CSR report, and it is easy to have endogenous doubts on these studies. The current research aims to discuss the effects of Taiwan's mandatory CSR report policy. The mandatory policy was unexpected and surprised those outside the firm. Mandatory disclosure and assurance can greatly reduce the reverse causality caused by endogenous doubts from previous researches. Therefore, it is worthwhile to use companies in Taiwan as research samples to discuss CSR-related issues. This paper was designed to use a random sample of external specifications for empirical evidence to provide empirical results with fewer interfering factors and to examine the regulatory effects of CSR report disclosure from voluntary to partial mandatory.

Second, the Taiwan market has its own characteristics. The company is mostly controlled by a family or a group. Therefore, its nature is different from other countries. Family-owned firms often possess excessive control rights beyond cash flow rights (Yeh, 2005) and family members dominate the top executive positions in the firm (Claessens et al., 2002). This could result in weaker shareholders' protection and inferior corporate governance mechanisms. The quality of corporate governance could also affect voluntary disclosure on CSR (Alfraih & Almutawa, 2017; Zhou, 2019). Therefore, Taiwan is essentially different from other countries. This study explored the impact of Taiwan's mandatory policy on CSR reporting in terms of investment-cash flow sensitivity. The results can be used as a reference for the management of CSR reports in many East Asian countries, especially those with many family-owned corporations.

2.4. Assurance service-related literature

Simnett and Nugent (2007) pointed out that the most important way to increase the trust of stakeholders in organizational commitment is to seek third-party certification or assurance. Moroney et al. (2012) found that the quality of the results for disclosure of an assurance environmental report was relatively high. Hodge et al. (2009) pointed out that the assured environmental report enhanced users' confidence towards this documentation. Quick and Inwinkl (2020) found that assurance positively influenced the confidence in CSR reports and that bankers were more likely to make favorable decisions toward reporting companies, such as approving credit applications, investing in the company, or recommending the purchase of shares to their clients. Stuart et al. (2020) found that the management's willingness to obtain independent assurance on disclosures was viewed as a positive ethical signal. This implies that assurance complements disclosure of CSR activities by adding protection against the impact of negative events. Under general recognition, several studies believed that assurance or audited information is more credible than that of non-assurance or unaudited information (Hodge, 2001; Pany & Smith, 1982). In summary, assurance is an important factor for upgrading the reliability of reports (Edgley et al., 2010).

From the above literature, if the independent CSR report is audited by an independent organization, credibility can be increased. Assurance services can be provided by accounting firms or third-party certification institutions. In the absence of mandatory legislative provisions, the scope of inspections, procedures, and the criteria for review by different organizational units will be relatively inconsistent.

The standards for policies revolving around international CSR reports are still in the stage of voluntary assurance. Kolk and Perego (2010) considered national institutional factors and found that companies in countries with a stakeholder orientation and weak governance mechanisms were likely to deliver CSR reports to be convinced. Compared with the international CSR report practice of voluntary assurance, Taiwan requires companies that meet the mandatory disclosure standards to compulsorily prepare CSR reports. To improve the quality of these reports, firms belonging to the food industry or those with no less than 50% of the company's revenue derived from food and beverage are required for mandatory assurance from an accountant. Taiwan's CSR report adopts partial mandatory assurance, which is different from other countries. At present, no literature discusses mandatory assurance policy for CSR report on firms' investment-cash flow sensitivity.

It can be inferred from the previous literature that CSR reports reveal non-financial information which can reduce information asymmetry between the company and the capital market. Under this situation, the assurance of independent third parties plays a role in improving the information quality being delivered to users, as well as reducing information asymmetry concerns between companies and users (Wallace, 2004), which could lessen potential agency problems and increase user confidence in CSR reports (Hodge et al., 2009). If the CSR report is convinced, it can amplify the benefits of the CSR report. This study explored whether the management policy for mandatory assurance for CSR reports can further alleviate the investment-cash

flow sensitivity of companies that are mandated to submit CSR reports to accountants for assurance. The following hypothesis is proposed:

H2. The mandatory assurance management policy for CSR reports helps to further decrease the investment-cash flow sensitivity of companies that are forced to take their CSR reports for accountant assurance.

3. Research models and variable definitions

3.1. Model 1

The first purpose of the study was to explore whether Taiwan's mandatory disclosure management policy for CSR reports helps alleviate the investment-cash flow sensitivity of companies.

Fazzari et al. (1988) used cash flow as a substitute variable for internal funds. The investment-cash flow sensitivity measures the dependence of a company's investment on internal funds. The lower the sensitivity of investment-cash flow, the lesser is its dependence on internal funds, which means lesser financial constraints in the firm. On the contrary, financially constrained companies, because of higher external capital costs, have investments that are more dependent on internal funds; thus, they are more sensitive to cash flow.

The research of Fazzari et al. (1988) led several studies to link investment-cash flow sensitivity to financial constraints (Ağca & Mozumdar, 2017; Alti, 2003; Brown & Petersen, 2009; Hovakimian, 2009). The reason for this situation is that there is a high gap between the cost of internal and external capital, and the access to external sources of capital is limited. The companies subject to financial constraints rely more on internally generated funds to raise money for investments. The empirical results of many papers supported the hypothesis that cash flow has a positive impact on investment (Ağca & Mozumdar, 2017; Brown & Petersen, 2009; Hovakimian, 2009).

Based on the argument of Fazzari et al. (1988), this paper examined whether the investment-cash flow sensitivity (financial constraint) of these compulsory companies are relaxed after the mandatory disclosure CSR report management policy. The empirical model is shown in Eq. (1).

$$INVEST_{i,t} = \beta_0 + \beta_1 CF_{i,t} + \beta_2 CF_{i,t} * CSRMD_{i,t} + \beta_3 CSRMD_{i,t} + \beta_4 Q_{i,t} + \beta_5 OUTPUT_{i,t} + \beta_6 PROFIT_{i,t} + \beta_7 AT_{i,t} + \varepsilon_{i,t} \quad (1)$$

where i denotes firms (i = 1, 2, …, N), t denotes year (t = 1, 2, …, T), INVEST is investment, CF is cash flow, CSRMD stands for CSR report mandatory disclosure management policy, CF*CSRMD is the interaction term between cash flow and CSRMD, Q stands for Tobin's Q, OUTPUT indicates firm output, PROFIT indicates profitability, AT is asset tangibility, and ε is an error term. If the coefficient of cash flow (CF) is positive and the coefficient of the interaction term (CF*CSRMD) between cash flow and pre-and-post mandatory disclosure policy dummy variable is negative, it means that the dependence of investment on cash flow is reduced after CSR report is forcibly disclosed. This implies that Taiwan's mandatory disclosure CSR report management policy can enhance information transparency of its capital market.

The descriptions and operational definitions of variables are as follows:

INVEST (investment): Based on previous literature (D'Espallier & Guariglia, 2015; Ding et al., 2013; Firth et al., 2012; Riaz et al., 2016), this article used the change in net fixed assets plus the depreciation expense for the year to measure investment.

CF (cash flow): Cash flow is the fund available within the company. It is an important indicator to measure the company's liquidity. If the company cannot obtain funds needed for investment at a low cost, the internal cash flow of the company may play an important role and be closely related to whether the company's investment project can be smoothly carried out. Referring to the practice of previous literature (Ağca & Mozumdar, 2017; Chen & Chen, 2012; Ding et al., 2013; Riaz et al., 2016), this paper defined cash flow as: the net income before extraordinary + depreciation + amortization.

CSRMD (CSR mandatory disclosure): CSRMD is a dummy variable. In this study, the period before the mandatory disclosure CSR report management policy was set to 0, and the period afterward was set to 1.

Q (Tobin's Q): Tobin's Q is the proxy variable for the company's growth opportunities. Classical investment theory expects Tobin's Q to be positively related to investment if it correctly measures the company's investment opportunities (Kumar et al., 2003). By definition, Tobin's Q is equal to the market value of the asset divided by the replacement cost of the asset. However, the replacement cost of the asset is not easy to measure. Therefore, this paper replaced Tobin's Q with the ratio of market value to book value as a proxy for the company's growth opportunities which was widely used in previous literature (Cleary, 1999; Kaplan & Zingales, 1997). This paper followed the practice of Baseri and Hakaki (2018) and defined it as: (Closing price x number share outstanding + total debt)/total assets.

OUTPUT (firm output): Output is the proxy variable of production to control the possible accelerator effect.

According to the expectation of acceleration theory, output has a positive impact on investment. Referring to previous literature (Ding et al., 2013; Riaz et al., 2016), this paper used the previous period's net sale as a proxy variable to measure output.

PROFIT (profitability): Myers (1984) pointed out that if the profitability of a company is high, the funds it receives should be sufficient to respond to the company's future investment, so the company's profitability will affect investment decisions. In the present study, it was measured by the net profit after tax for the period divided by the net sales.

AT (asset tangibility): Almeida and Campello (2007) believed that the real assets available for secured mortgages will have better borrowing power, and the better borrowing capacity will enable the company to make further investments. This study measured AT as fixed assets divided by the total assets.

To eliminate the effects of scale, the current paper referred to previous literature (Ding et al., 2013; Fazzari et al., 1988; Hoshi et al., 1991; Riaz et al., 2016) for the measurement scales. INVEST (investment), CF (cash flow), and OUTPUT (firm output) were normalized using the company's initial capital stock. With reference to the definition of Cleary (1999), the capital stock was measured by the company's net fixed asset value at the beginning of the period in this study.

Since Taiwan's mandatory disclosure regulations on CSR reports were first implemented in 2014, the 2014 CSR reports were released in 2015. Therefore, all CSR reports submitted in 2014 were voluntary disclosures. This study used the CSR report disclosure data from the Taiwan Stock Exchange Corporation's (TWSE) Market Observation Post System to select the sample. The following inclusion criteria were used to select the samples for this study: (1) it is classified as a firm subjected to mandatory disclosure by policy regulations; and (2) it has never disclosed any CSR reports before the mandatory disclosure policy. The purpose of the inclusion criteria is to control the samples under the same CSR mandatory disclosure mechanism to avoid interference from other factors.

This study compared the companies that issued CSR reports in 2015 (262 companies) with those that issued in 2014 (126 companies) to select the samples that should disclose CSR reports in accordance with the CSR report mandatory disclosure regulations (101 companies). Because the banking and insurance industry has several significant differences in terms of industrial characteristics and accounting systems, this study deleted 13 companies belonging to the banking and insurance industry. Finally, 3 companies were deleted with incomplete data. After the sample selection process, a total of 85 companies were included in the study.

Since Taiwan's mandatory disclosure regulation on CSR reports was issued in 2014, the study period included pre-and-post mandatory disclosure CSR report management policy (from 2012 to 2018). The annual data used for analysis consisted of 595 observations, which was obtained by multiplying the number of sample companies (N = 85) to the number of years of the research period (N = 7). Data were obtained from a database maintained by the Taiwan Economic Journal (TEJ), a leading credit analysis research agent. In this paper, financial information was taken from the TEJ Finance database and the stock price was taken from the TEJ Equity database.

3.2. Model 2

The second purpose of this study was to explore whether Taiwan's mandatory assurance management policy for CSR reports can further alleviate the investment-cash flow sensitivity of companies that are forced to subject their reports to accountant assurance. The empirical model is shown in Eq. (2).

$$INVEST_{i,t} = \delta_0 + \delta_1 CF_{i,t} + \delta_2 CF_{i,t} * MA_{i,t} + \delta_3 MA_{i,t} + \delta_4 Q_{i,t} + \delta_5 OUTPUT_{i,t} + \delta_6 PROFIT_{i,t} + \delta_7 AT_{i,t} + \varepsilon_{i,t} \quad (2)$$

where i denotes firms (i = 1, 2, …, N), t denotes year (t = 1, 2, …, T), INVEST is investment, CF is cash flow, MA denotes CSR report mandatory assurance management policy, CF*MA is the interaction term between cash flow and MA, Q stands for Tobin's Q, OUTPUT indicates firm output, PROFIT indicates profitability, AT is asset tangibility, and ε is an error term.

In this study, MA refers to mandatory assurance management policy. If a company is forced to assure its CSR report by an accountant under Taiwan's CSR report mandatory assurance management policy, MA was set to 1, otherwise it was set to 0. Assurance service can also be done by a third-party rather than an accountant. This study employed the robustness test to compare the effects of assurance by an accountant and by third-party organization. The remaining variables are the same as Eq. (1). If the coefficient of cash flow (CF) is positive and the coefficient of the interaction term between cash flow and mandatory assurance dummy variable (CF*MA) is negative, then this signifies that the dependence of investment on cash flow of mandatory accountant assurance firms is less than others. This implies that Taiwan's mandatory accountant assurance CSR report management policy can improve capital market's information transparency.

The second purpose of this paper was to investigate the impact of the mandatory accountant assurance CSR report management policy on firms' investment-cash flow sensitivity, i.e., to review the effectiveness of the mandatory assurance of CSR report management policy. The sample is similar to the first hypothesis, but the study period only included the implementation period of the mandatory disclosure CSR report management policy (i.e., 2015 to 2018) to control the effect of the report management policy. Moreover, this paper excluded firms that had voluntary accountant assurances to exclude the effect of voluntary assurance. Among the sample firms (85 firms), 30 firms' CSR reports were assured by accountants (25 were mandatory and 5 were voluntary), 16 firms' CSR reports were submitted to a third-party organization for assurance, and 39 firms' CSR reports did not undergo assurance.

4. Empirical results and discussions

4.1. Descriptive statistics

Descriptive statistics for the variables are shown in Table 1. The sample consisted of 85 firms, and the study period was between 2012 and 2018. The observation was 595. The means for INVEST (investment), CF (cash flow), Q (Tobin's Q), OUTPUT (firm output), PROFIT (profitability), and AT (asset tangibility) were 0.087, 2.746, 0.921, 17.159, 0.143, and 0.224, respectively. The variable with the largest degree of dispersion was OUTPUT, and the variable with the smallest was AT.

4.2. Empirical results

The empirical results are shown in Table 2. Panel A presents the results of model 1. The cash flow coefficient was 0.054 and the level of significance was at 5%, suggesting that firms rely on cash flow for financing their investments. The coefficient of interaction term (CF*CSRMD), the focus of the present analysis, was −0.013 and the level of significance was at 10%. The negative interaction term between cash flow and the pre-and-post mandatory disclosure policy dummy variable indicates that it helped decrease the reliance of investment on cash flow of the mandatory disclosure CSR report firms. The investment-cash flow sensitivity ranged from 0.054 to 0.041 after the effect of the mandatory disclosure policy was taken into consideration. The empirical result supports H1; that is, the mandatory disclosure management policy for CSR reports helps in alleviating the investment-cash flow sensitivity of companies that are forced to disclose their CSR reports. A decrease in the investment-cash flow sensitivity after the implementation of the mandatory disclosure policy indicates that there is a decrease in information asymmetry and agency problems in companies that were forced to disclose their CSR reports. This decrease implies that the policy of interest does have a positive influence on increasing the information transparency of its capital market.

The results for the other control variables are as follows: CSRMD was positive which indicates that the investment increased after the mandatory disclosure of CSR reports policy; however, it was not significant. The Tobin's Q coefficient was positive as expected but it was not significant at the 10% level. The OUTPUT coefficient was significant and negative at the 1% level which did not meet the expectation of the acceleration theory. This may be because the company considers the demand for longer-term sales forecasts when investing, rather than just being affected by the previous period's output. The PROFIT coefficient was positive as expected but it was not significant at the 10% level. As predicted, the coefficient of AT was significantly positive at the 5% level which was consistent with that of previous research (Almeida & Campello, 2007).

Table 1
Descriptive statistics.

	INVEST	CF	Q	OUTPUT	PROFIT	AT
Mean	0.087	2.746	0.921	17.159	0.143	0.224
Maximum	36.403	240.868	7.527	885.744	16.739	0.819
Minimum	−0.985	−24.225	−0.060	0.032	−8.573	0.0004
Std. Dev.	1.572	17.285	0.795	66.718	0.940	0.169
Observations	595	595	595	595	595	595
Cross sections	85	85	85	85	85	85

Note: INVEST: investment; CF: cash flow; Q: Tobin's Q; OUTPUT: firm output; PROFIT: profitability; AT: asset tangibility.

Table 2
Results of investment model.

Panel A: Model 1			Panel B: Model 2		
Variable	Coefficient	P-Value	Variables	Coefficient	P-Value
CF	0.054**	0.013	CF	0.166***	<0.001
CF*CSRMD	−0.013*	0.087	CF*MA	−0.156***	<0.001
CSRMD	0.162	0.203	MA	0.043	0.810
Q	0.219	0.227	Q	0.127	0.249
OUTPUT	−0.008***	<0.001	OUTPUT	−0.013***	<0.001
PROFIT	0.030	0.692	PROFIT	−0.144**	0.043
AT	3.221**	0.027	AT	−0.075	0.850
A-R^2	0.28		A-R^2	0.48	
Obs.	595		Obs.	320	

Notes: (a) Dependent variable: INVEST (investment). (b) CF: cash flow; CSRMD: CSR report mandatory disclosure management policy; MA (mandatory assurance): CSR report mandatory assurance management policy; Q: Tobin's Q; OUTPUT: firm output; PROFIT: profitability; AT: asset tangibility. (c) ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. (d) The study period of model 1 was from 2012 to 2018 and the total sample was 85; the number of observations was 595. (e) In model 2, 5 voluntary accountant assurance firms were excluded; the study period of model 2 was from 2015 to 2018 and the total sample was 80; the number of observations was 320. (f) Constant term and the individual effects are not shown in this table.

Panel B of Table 2 shows the results of model 2. A company that was forced to take its CSR report to an accountant for assurance was given an MA value of 1. There were 25 firms with an MA of 1 and 5 firms that had voluntary accountant assurance were excluded from the study. The MA of the remaining 55 firms were set at 0. The cash flow coefficient was 0.166 and the level of significance was at 1%. The coefficient of interaction term (CF*MA) was −0.156 and the level of significance was at 1%. The negative interaction between cash flow and MA dummy variable indicates that submission of CSR reports for accountant assurance helped decrease the reliance of investments on cash flow than those firms with non-accountant assurance or without assurance at all. The above finding supports H2; that is, the mandatory assurance management policy for CSR reports helps to further decrease the investment-cash flow sensitivity of companies that are forced to take their CSR reports for accountant assurance. A decrease in the investment-cash flow sensitivity after mandatory accountant assurance of CSR reports implies a decrease in information asymmetry and agency problems of companies that were forced to take their CSR reports for accountant assurance after Taiwan implemented the partial mandatory assurance policy. This signifies that the said policy may improve the information transparency of its capital market.

The results for the other control variables are as follows: MA was positive which indicates that the investment increased after the mandatory assurance of CSR reports by accountants

but this increase was not significant. The results of Tobin's Q and OUTPUT were the same as model 1. PROFIT was negative and significant and AT was negative but not significant. The study periods for models 1 and 2 were not the same; thus, the results were not the same and inconsistent with each other.

4.3. Robustness tests

4.3.1. Controlling for the industrial effects

Article 2 of Taiwan's policy entitled "Taiwan Stock Exchange Corporation Rules Governing the Preparation and Filing of Corporate Social Responsibility Reports by TWSE Listed Companies" stipulates the scope of listed companies that should prepare CSR reports. The first of these points out that at the end of the most recent fiscal year, companies belonging to the food industry, chemical industry, and financial and insurance industry should prepare a CSR report. Since financial and insurance companies are not within the scope of this research, the industry effects were controlled by adding two industry dummy variables namely: IND1 for the food industry which was set to 1, with the rest set to 0; and IND2 for the chemical industry which was also set to 1, with the rest set to 0. As shown in Table 3, the results were unchanged, both H1 (the mandatory disclosure management policy for CSR reports helps in alleviating the investment-cash flow sensitivity of companies that are forced to disclose their CSR reports) and H2 (the mandatory assurance management policy for CSR reports helps to further decrease the investment-cash flow sensitivity of companies that are forced to take their CSR reports for accountant assurance) were supported after controlling for the industry effects. In sum, Taiwan's CSR report management policy helped decrease the reliance of investment on cash flow which means the policy could increase the information transparency of its capital market. The results for other control variables were the same after controlling industrial effects. IND2 was significantly positive which means that the chemical industry's investment was greater than other industries in the sample.

Table 3
Results of investment model-controlling for the industrial effects.

Panel A: Model 1			Panel B: Model 2		
Variable	Coefficient	P-Value	Variables	Coefficient	P-Value
CF	0.054**	0.014	CF	0.168***	<0.001
CF*CSRMD	−0.013*	0.088	CF*MA	−0.157***	<0.001
CSRMD	0.163	0.203	MA	0.004	0.984
Q	0.218	0.230	Q	0.074	0.519
OUTPUT	−0.008***	<0.001	OUTPUT	−0.013***	<0.001
PROFIT	0.030	0.691	PROFIT	−0.140*	0.053
AT	3.276**	0.026	AT	−0.327	0.444
IND1	0.030	0.978	IND1	0.233	0.362
IND2	−0.267	0.735	IND2	0.309*	0.094
A-R^2	0.28		A-R^2	0.48	
Obs.	595		Obs.	320	

Notes: (a) Dependent variable: INVEST (investment). (b) CF: cash flow; CSRMD: CSR report mandatory disclosure management policy; MA (mandatory assurance): CSR report mandatory assurance management policy; Q: Tobin's Q; OUTPUT: firm output; PROFIT: profitability; AT: asset tangibility. IND1: food industry was set to 1, the rest was set to 0; IND2: chemical industry was set to 1, and the rest was set to 0. (c) ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. (d) The study period of model 1 was from 2012 to 2018 and the total sample was 85; the number of observations was 595. (e) In model 2, 5 voluntary accountant assurance firms were excluded. The study period of model 2 was from 2015 to 2018 and the total sample was 80; the number of observations was 320. (f) Constant term and the individual effects are not shown in this table.

4.3.2. The comparisons between group-controlled firms and non-group-controlled firms

Previous literature pointed out that many companies in Taiwan are controlled by groups of families. For instance, Lemmon and Lins (2003) and Filatotchev et al. (2005) noted that Taiwan's market is mainly composed of family-owned businesses with family members as top executives. In order to increase the robustness, this study further divided the sample into group-controlled firms and non-group-controlled firms to determine if this factor affects the results of the study.

The study's definition of a group-controlled firm was based on the Taiwan Economic Journal's (TEJ) group composition database which identifies group firms as co-governance, single-family dominance, public equity dominance, and professional manager governance. Among the 85 sample companies in this study, 72 are group-controlled, of which, 56 are controlled by a single family, and 16 are controlled by other controllers (co-governance, public equity dominance, and professional manager governance); the remaining 13 companies are non-group-controlled. This shows that most of the sample companies are mainly controlled by a group or a family, which is consistent with those of previous literature.

The empirical results for models 1 and 2 are shown in Panels A and B of Table 4. In Panel A, both groups' CF coefficients were significant and positive and the CF*CSRMD coefficients were significant and negative, which means that both groups' investment-cash flow sensitivity decreased after the CSR report mandatory disclosure policy. H1 is still supported by the empirical result. In Panel B, both groups' CF coefficients were significant and positive and the CF*MA coefficients were significant and negative, which means that both groups' investment to cash flow sensitivity also decreased after the CSR report mandatory assurance policy. H2 is still supported by the empirical result.

The results of this study remained unchanged and were not affected by the type of control. Group or family control may affect the quality of corporate governance, and the quality of corporate governance may affect voluntary disclosure of CSR reports (Alfraih & Almutawa, 2017; Zhou, 2019). This study mainly discussed the effects of Taiwan's mandatory policy on CSR report which is an exogenous factor and is not affected by the quality of corporate governance under group or family control. Therefore, our results were unchanged after considering if the firms were group controlled or not.

4.3.3. The comparisons between accountant assurance and other third-party assurance

The second purpose of this research paper was to discuss whether the mandatory assurance management policy of the CSR report can further reduce the investment to cash flow

Table 4
Results of investment model-group and non-group companies.

Panel A: Model 1				
Variable	group companies (72 firms)		non-group companies (13 firms)	
	Coefficient	P-Value	Coefficient	P-Value
CF	0.026**	0.037	2.603***	<0.001
CF*CSRMD	−0.001*	0.087	−0.981***	<0.001
CSRMD	0.036***	<0.001	−0.226***	<0.001
Q	0.070***	<0.001	−0.061	0.407
OUTPUT	−0.001	0.163	−0.457***	<0.001
PROFIT	0.003	0.375	−3.319***	<0.001
AT	0.745***	<0.001	−1.427***	0.003
A-R^2	0.34		0.88	
Obs.	504		91	
Panel B: Model 2				
	group companies (67 firms)		non-group companies (13 firms)	
Variable	Coefficient	P-Value	Coefficient	P-Value
CF	0.154***	<0.001	7.368***	<0.001
CF*MA	−0.147***	<0.001	−1.629*	0.076
MA	0.058	0.756	0.193	0.260
Q	−0.033	0.795	0.195*	0.053
OUTPUT	−0.008***	<0.001	−0.326***	<0.001
PROFIT	−0.133**	0.046	−8.430***	<0.001
AT	−0.475	0.376	−2.401***	<0.001
A-R^2	0.47		0.76	
Obs.	268		52	

Notes: (a) Dependent variable: INVEST (investment). (b) CF: cash flow; CSRMD: CSR report mandatory disclosure management policy; MA (mandatory assurance): CSR report mandatory assurance management policy; Q: Tobin's Q; OUTPUT: firm output; PROFIT: profitability; AT: asset tangibility. (c) ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. (d) The study period of model 1 was from 2012 to 2018 and the study period of model 2 was from 2015 to 2018. (e) In model 2 of group companies, 5 voluntary accountant assurance firms were excluded. (f) Constant term and the individual effects are not shown in this table.

sensitivity of the CSR reporting companies. Pflugrath et al. (2011) pointed out that CSR messages are more credible when they are convinced and authenticated. They found that in the United States, such guarantees are most valuable when done by professional accountants. The present study further compared the effects of accountant assurance with third-party assurance; thus, samples that did not submit their CSR report for assurance (39 firms) were not included. The model used was the same as Eq. (2), except that the MA dummy variable was replaced by the AA dummy variable. If a company hired an accountant for CSR report assurance, the AA was set to 1 (30 firms); if the CSR report was taken to a third-party, the AA was set to 0 (16 firms). The analysis of model (2) was re-examined to compare the effects of accountant assurance with third-party assurance.

The results are shown in Table 5. In Panel A, the cash flow coefficient was 0.175 and the level of significance was at 1%; the coefficient of interaction term (CF*AA) was −0.165 and the level of significance was at 1%. This signifies that the investment-cash flow sensitivity was lower for companies

Table 5
Results of investment model-comparisons between accountant assurance and other third-party assurance.

Panel A			Panel B		
Variable	Coefficient	P-Value	Variable	Coefficient	P-Value
CF	0.175***	<0.001	CF	0.176***	<0.001
CF*AA	−0.165***	<0.001	CF*AA	−0.166***	<0.001
AA	0.201	0.477	AA	0.231	0.493
Q	−0.020	0.908	Q	−0.030	0.862
OUTPUT	−0.013***	<0.001	OUTPUT	−0.014***	<0.001
PROFIT	−0.017	0.866	PROFIT	−0.018	0.855
AT	−0.216	0.764	AT	−0.359	0.627
			IND1	−0.022	0.946
			IND2	0.461	0.272
A-R^2	0.54		A-R^2	0.54	
Obs.	184		Obs.	184	

Notes: (a) Dependent variable: INVEST (investment). (b) CF: cash flow; AA: account assurance; Q: Tobin's Q; OUTPUT: firm output; PROFIT: profitability; AT: asset tangibility. IND1: food industry was set to 1, the rest was set to 0; IND2: chemical industry was set to 1, and the rest was set to 0. (c) ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. (d) The study period was from 2015 to 2018 and the sample firm was 46; the number of observations was 184. (e) Constant term and the individual effects are not shown in this table.

that had their CSR reports assured by an accountant than those by a third-party. Panel B shows the results after controlling for the industrial effects. The cash flow coefficient obtained was 0.176 and the level of significance was at 1%. The coefficient of interaction term (CF*AA) was −0.166 and the level of significance was at 1%. The results revealed robustness; CSR reports authenticated by accountants have a lower reliance of investment on cash flow than those done by a third-party.

5. Conclusion

The management policy on Taiwan's CSR report encourages voluntary disclosure to partial mandatory disclosure. Apart from the mandatory disclosure, some companies are further requested to deliver CSR reports certified by an accountant for mandatory assurance, thereby leading to a more effective status in the capital markets.

This study found that Taiwan's capital market may be a suitable environment for CSR management policy research. This paper may serve as a benchmark for further studies regarding this policy. At present, most of the literature on CSR-related issues focused on either voluntary disclosure or voluntary assurance. Therefore, academically, this paper is expected to have incremental contributions to the field of CSR practice in Taiwan's management policy implementation.

Using 85 firms listed on the Taiwan Stock Exchange as samples, the results suggested that firms' investment-cash flow sensitivity decreased after mandatory CSR report disclosure. This implies that the partial mandatory CSR report disclosure policy relaxes firms' investment-cash flow sensitivity. Moreover,

it was found that the investment-cash flow sensitivity was lower for firms that had their CSR reports certified by an accountant than those by a third-party or those without assurance. The results also revealed that the management policy requiring partial mandatory assurance for CSR reports decreased firms' investment-cash flow sensitivity. This indicates that Taiwan's partial CSR report management policies help to decrease the investment-cash flow sensitivity of mandatory firms, which is evidence for increasing information transparency in its capital market after the partial mandatory disclosure and mandatory assurance CSR report management policies.

Our findings have several policy implications. First, since the Taiwan stock market is an emerging capital market, improving its information transparency to attract funds is an important issue that needs to be discussed. The results of this article supported mandatory disclosure of CSR reports and mandatory assurance of CSR reports. It was found that both can help alleviate the problem of financial constraints. This means that the CSR management policy is conducive to enhancing the transparency of capital market information. However, only some companies are required to disclose or ensure their CSR reports. Therefore, the author suggests that the management authority may conduct more research and expand the scope of mandatory disclosure and assurance of CSR reports. Second, as far as companies are concerned, only some are required to disclose and ensure their CSR reports. However, under the goal of sustainable development, the expanded implementation of the mandatory disclosure and assurance of CSR reports policies are foreseeable. In the face of this future trend, the company may be able to respond early and think about the company's CSR management, disclosure, and assurance policies. Finally, the research results of this article can be used as a reference for other emerging capital markets when formulating their CSR management policies.

Conflict of interest

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or nonfinancial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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